

EvalVitals

The model health platform
connecting AI to
real-world applications.

Evaluate. Diagnose. Repair.
Generate data. Verify.

INCUBATED BY

NEBIUS RESEARCH STUDIO



Evals end where the expensive work begins.

A good score tells you something is wrong.
It doesn't tell you why, or how to fix it.

Evaluation

Overall score

67%

	Factuality	72%
	Reasoning	61%
	Safety	83%
	Task completion	57%



But you still need to find:



Why did it fail?



What is the mechanism?



How do we fix it?



Did the fix work?



Will it generalize?

From frontier models to real-world impact.

EvalVitals connects model capabilities to reliable application performance.



Model debugging still scales with people.

When a model fails, teams spend days or weeks to:



Reproduce the failure



Inspect and analyze



Hypothesize causes



Try fixes: prompts, data, training



Retest and iterate

*Same failure.
Different prompt?
More data?
Full fine-tune?*

*Brilliant people.
Still a manual loop.*

EvalVitals automates the model health loop.

Evaluation, diagnosis, repair and training data in one verified loop.

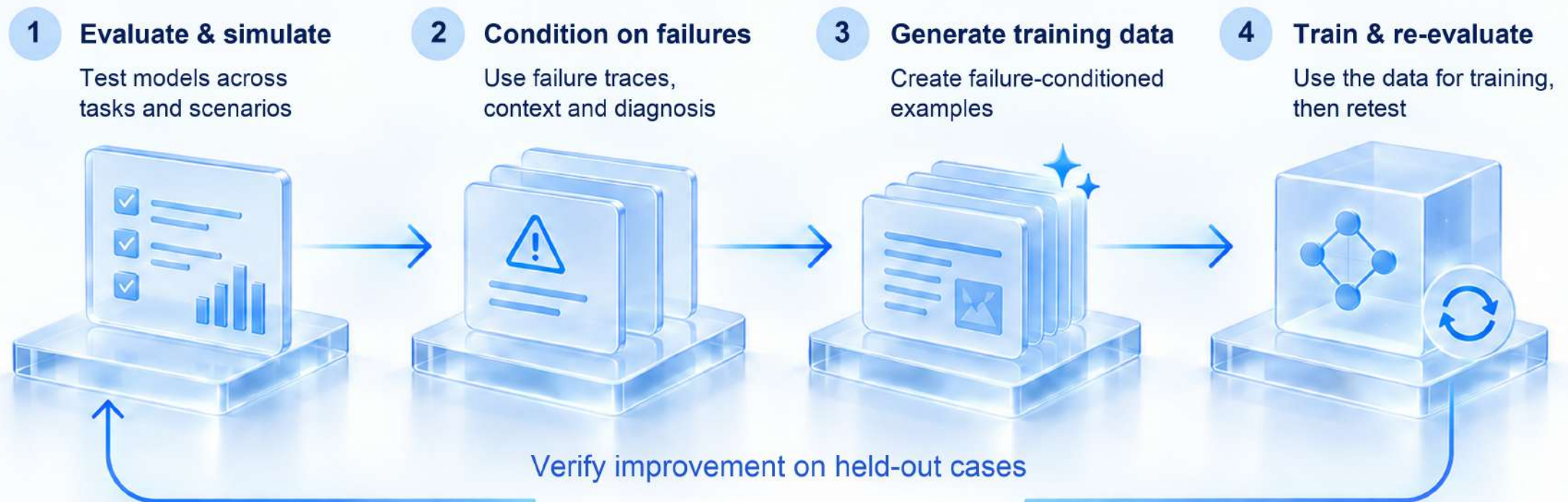


One model health platform. Many models, modalities and tasks.



Evaluation drives training-data generation

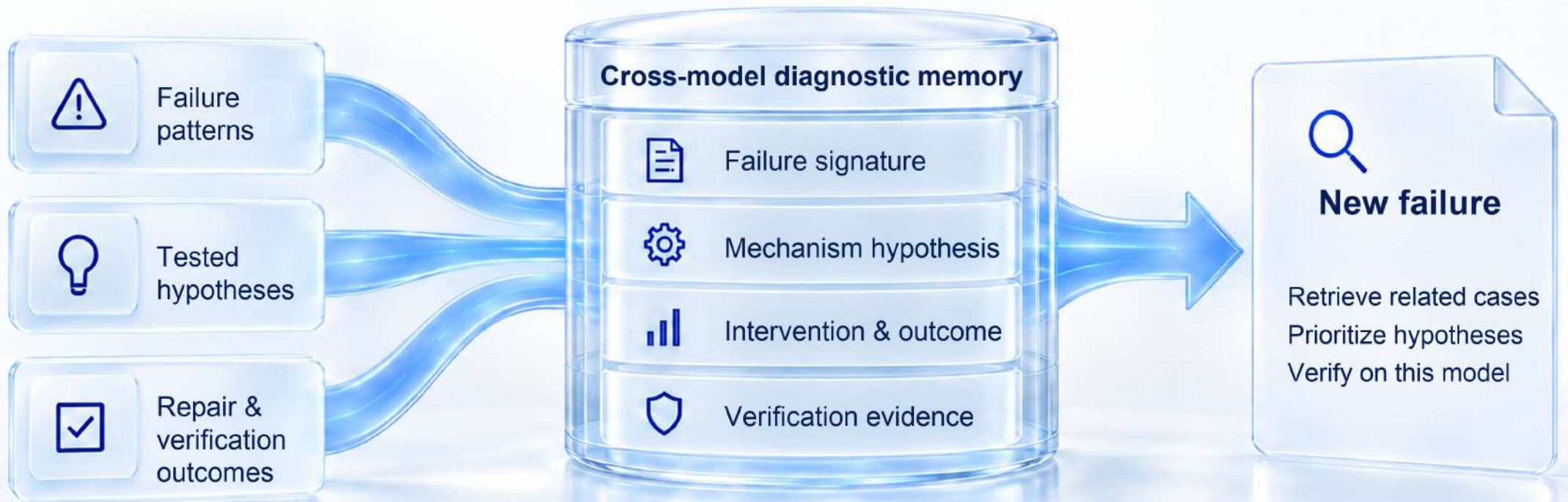
Large-scale simulation tests expose failures and guide what data to generate.



Output: failure-conditioned data for further model training

A shared memory of verified diagnoses

Alongside training data, EvalVitals accumulates empirical diagnostic experience.



Past evidence guides the diagnosis. Current tests verify it.

Training data and diagnostic memory compound together.



A more capable, more reliable world.



Enterprises
Deploy AI confidently



Government
Safe and responsible AI



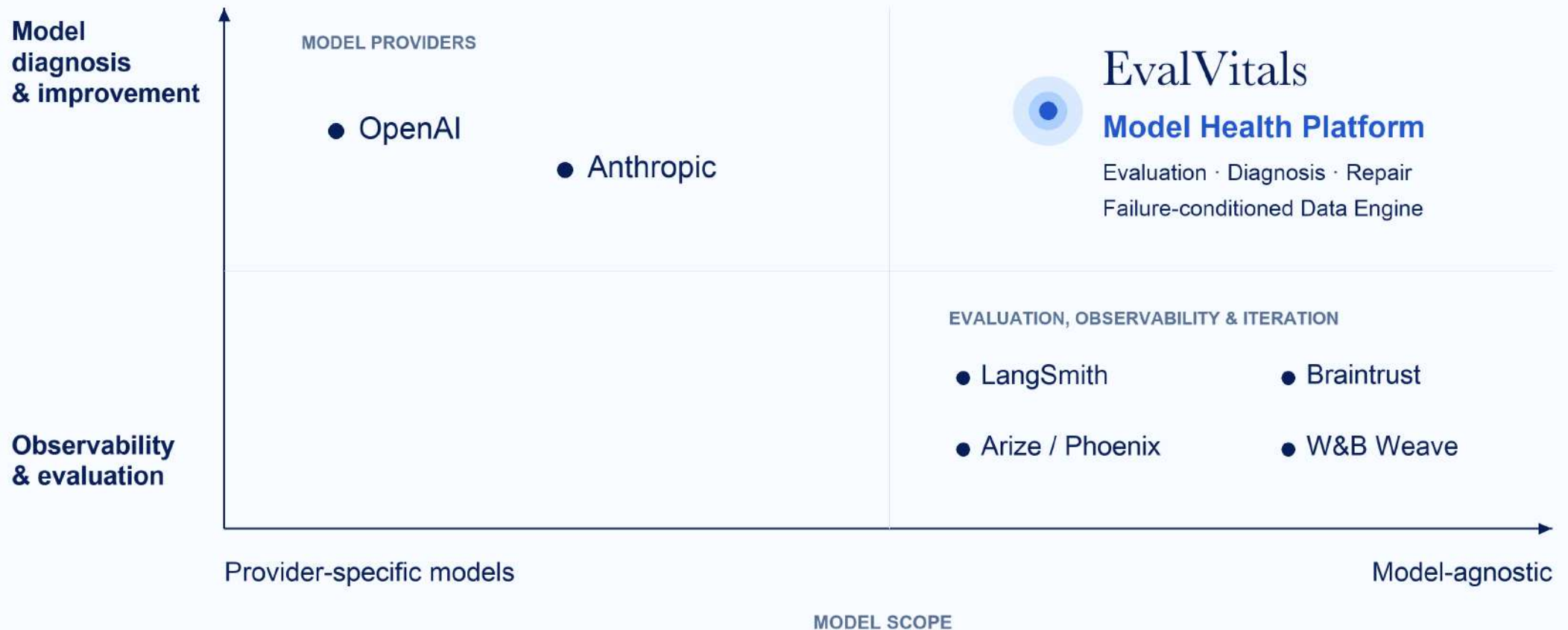
Science & Industry
Accelerate discovery



Society
Solve real-world problems

A model health platform across the improvement cycle

Our positioning: evaluation, diagnosis, repair and a failure-conditioned data engine.



A platform business with compounding data assets

1. Model Health Runs

- Evaluation & simulation
- Diagnosis and repair
- Usage-based + compute
- Model teams / FDEs



START TODAY

2. Enterprise Platform

- Continuous monitoring
- Team workflows
- Annual subscription + usage
- Enterprises / government



SCALE WITH CUSTOMERS

3. Data & Intelligence

- Failure-conditioned data
- Cross-model diagnostic memory
- Verification evidence
- Data and platform licensing



COMPOUND OVER TIME



EvalRx: released software, reported results

Open-source diagnosis and repair with an auditable evidence trail.

TEAM-REPORTED RESEARCH RESULTS

12 / 12

settings improved

+5.8 pp

average gain

+1.4–10.9 pp

gain range

Across evaluated model–modality settings.

Source: team technical report abstract, September 2026.

Python package

EvalRx 0.1.2

`pip install evalrx`

pypi.org/project/evalrx/

Interactive reports

VLM and audio runs

Hypotheses, tests
and repair outcomes

evalvitals.github.io/evalrx/demo/

Technical report

TECHNICAL REPORT

September 2026

EvalRx

Closing the Loop from Model Evaluation to Auditable Repair Recipes

An agentic system for auditable diagnosis and repair across modalities

Eval Vitals Team

ABSTRACT

Model evaluation reveals *where* models fail, but rarely *why* or whether a proposed fix genuinely works. We introduce EVALRX, an open-source agentic framework for statistically grounded model diagnosis and repair. EVALRX discovers systematic failure modes, explores candidate mechanisms, formulates falsifiable hypotheses, and verifies them by executing frozen, pre-specified tests before using supported diagnoses to guide tiered repair. Candidate repairs are validated against the unchanged model and promoted only when they improve the target behavior without unacceptable regression. Across language, vision-language, and audio-language settings spanning Gemini, Qwen, Gemma, and Nemotron, EVALRX improves performance in all 12 evaluated model-modality settings, with gains ranging from +1.4 to +10.9 percentage points (+5.8 points on average). These results show that model evaluation can serve not only to measure failures, but to support an auditable process for diagnosing and repairing them.

EvalRx technical report · September 2026

Published demo runs show the audit workflow.

Demo runs are separate from the aggregate research results.

Research depth. Experience building real systems.



Xiaoxiao Li

UBC PROFESSOR · GOOGLE GENAI

CIFAR AI Chair and Canada
Research Chair at UBC.

Research in trustworthy ML,
optimization, LLMs and AI agents.



Ruinan Jin

UBC · AWS & GOOGLE

Graduate student at UBC.

Experience building customer-facing
systems at AWS and Google.



Jiaqi Liu

UBC · FORMER ALIBABA RESEARCHER

Graduate student at UBC.

Worked on the Ovis open-source
model and Qwen.

Making advanced
intelligence reliable
in real-world applications.

Evaluate. Diagnose. Repair.
Generate data. Verify.

EvalVitals

A NeoLab incubated by Nebius Research Studio.

